

House Price Prediction Project

Insights and Summary

Key observations:

Larger houses generally had higher prices.

- Houses with more bathrooms tended to be more expensive.
- Air conditioning and better furnishing increased property value.
- Features such as hot water heating and guest rooms had relatively smaller effects.

Interesting Findings

1. Random Forest, which is often more powerful than Linear Regression, performed worse on this dataset.
2. This suggests that the relationship between house features and price is relatively linear.
3. The dataset contained no missing values and no duplicate records, making data preparation straightforward.
4. Area had a much stronger influence than most other features.

Final Conclusion

This project successfully demonstrated the use of machine learning for house price prediction. After preprocessing the data and converting categorical variables into numerical form, both Linear Regression and Random Forest models were trained. Linear Regression achieved the best performance with an R^2 score of approximately 65%, indicating a moderate level of predictive accuracy. The analysis showed that property area, bathrooms, parking, and air conditioning are among the most important factors influencing house prices. With additional data and feature engineering, the model's accuracy could be improved further, making it a useful tool for real estate pricing and decision making.